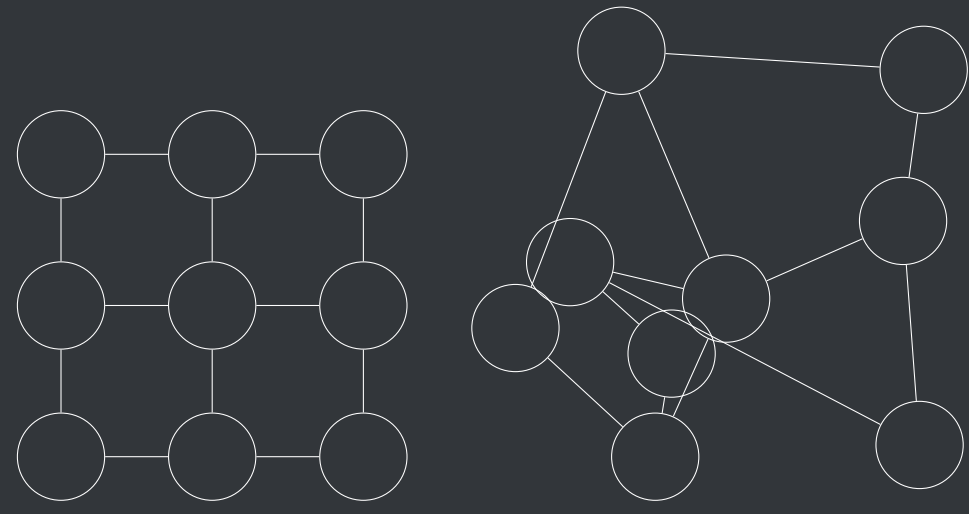


# Neighborhood-Preserving Translations on Graphs

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## 1. Motivation



**Figure:** Example of the same graph depicted twice with different vertices locations, emphasizing why it is not trivial to define translations on graphs that match usual ones on regular domains.

## 2. Related Work

- Laplacian matrix of the graph, defined as  $\mathbf{L} = \mathbf{D} - \mathbf{W}$  -  $\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$  -  $\hat{\mathbf{x}} \triangleq \mathbf{U}^\top \mathbf{x}$  graph Fourier transform -  $\mathbf{U}(\mathbf{U}^\top \mathbf{x}_1 \odot \mathbf{U}^\top \mathbf{x}_2)$  a translation is not defined by a “shift” but by a destination vertex : Shuman
- isometric operator with respect to the  $\ell_2$ -norm. The translation of a signal they propose consists in multiplying the signal by some matrix which is an exponential of an imaginary diagonal matrix : Girault
- some definitions of translation (or “shift”) exist for powers of a ring graph : Ekambaram

## 3. Proposed Definitions

### Definition (*Perfect graphical translation*)

We say that an application  $f$  from  $\mathcal{V}$  to  $\mathcal{V}$  is a perfect graphical translation of  $\mathcal{G}$  if:

- 1  $f$  is bijective;
- 2 for all vertex  $v$  in  $\mathcal{V}$ ,  $f(v)$  is a neighbor of  $v$ ;
- 3 for all couple  $(v_1, v_2) \in \mathcal{V}^2$ ,  $(v_1, v_2) \in \mathcal{E} \Leftrightarrow (f(v_1), f(v_2)) \in \mathcal{E}$ .

### Definition (*Candidate graphical translation*)

We say that an application  $f$  from  $\mathcal{V}$  to  $\mathcal{V} \cup \{\omega\}$  is a candidate graphical translation if:

- 1  $f|_{\mathcal{V}_{0,f}}$  is injective;
- 2 for all vertex  $v$  in  $\mathcal{V}_{0,f}$ ,  $f(v)$  is a neighbor of  $v$ ;
- 3 for all couple  $(v_1, v_2) \in \mathcal{V}_{0,f}$ ,  $(v_1, v_2) \in \mathcal{E} \Leftrightarrow (f(v_1), f(v_2)) \in \mathcal{E}$ .

### Definition (*Generalized graphical translation*)

We say that an application  $f$  from  $\mathcal{V}$  to  $\mathcal{V} \cup \{\omega\}$  is a (generalized) graphical translation if:

- 1  $f$  is a candidate graphical translation;
- 2  $\forall v \in \mathcal{V}_{0,f}$ , for every  $g$  candidate graphical translation such that  $g(v) = f(v)$ ,  $|\mathcal{V}_{0,g}| \leq |\mathcal{V}_{0,f}|$ .

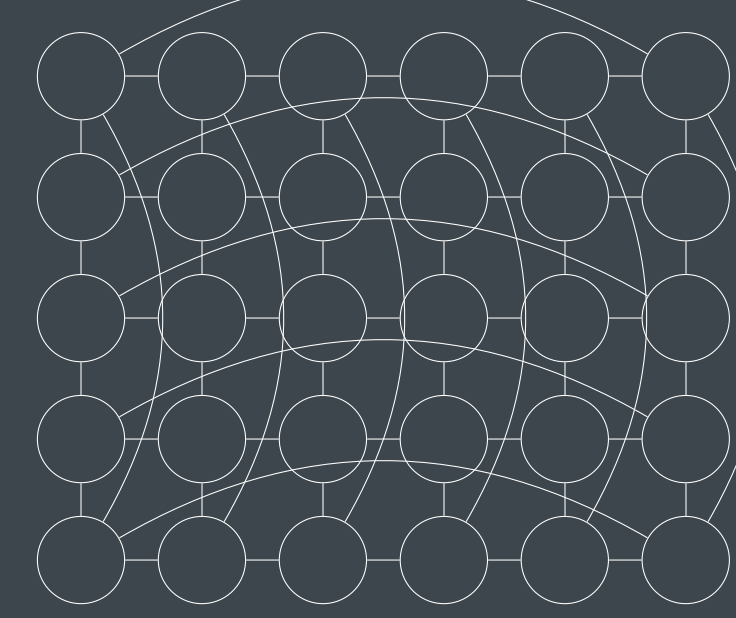
### Definition (*Geometrical translation*)

We say that an application  $f$  from  $\mathcal{V}$  to  $\mathcal{V} \cup \{\omega\}$  is a geometrical translation on a grid graph  $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$  if there is  $\delta = \pm \mathbf{e}_i$  such that

$$\forall v \in \mathcal{V}, f(v) = \begin{cases} v + \delta & \text{if } v + \delta \in \mathcal{V} \\ \omega & \text{otherwise} \end{cases}.$$

## 4. Analysis on grid graphs

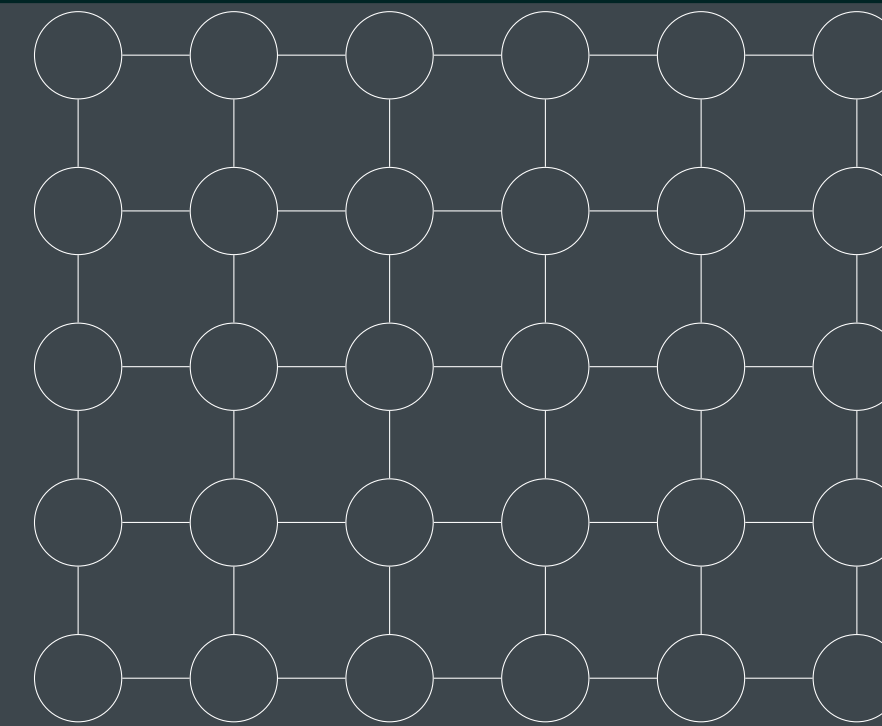
### 4.1. Analysis on cyclic grid graphs



**Proposition :** Let  $f$  be a perfect graphical translation on a cyclic grid graph, then  $f$  is a graphical translation.

**Proof :** Let us denote  $\delta = f(\mathbf{0}) - \mathbf{0} = f(\mathbf{0})$  and let  $\mathbf{v} \in \mathcal{V}$ . As a cyclic grid graph is connected, there is a path from  $\mathbf{0}$  to  $\mathbf{v}$ . By propagating the contamination lemma along the path, neighbor after neighbor, we obtain that  $f(\mathbf{v}) = \mathbf{v} + \delta$ .

### 4.2. Analysis on noncyclic grid graphs



Let us consider a noncyclic grid graph with parameters  $\mathbf{d}$  and  $(\ell_i)_{1 \leq i \leq d}$  such that:

$$\forall i \in \llbracket 1, d-1 \rrbracket, \ell_i \geq (2 \prod_{k=i+1}^d \ell_k) + 2.$$

Moreover, we force  $\ell_d \geq 3$ .

Let us first point out that the function  $f_\omega$ :

$$f_\omega : \begin{cases} \mathcal{V} \rightarrow \mathcal{V} \cup \{\omega\} \\ v \mapsto \omega \end{cases}$$

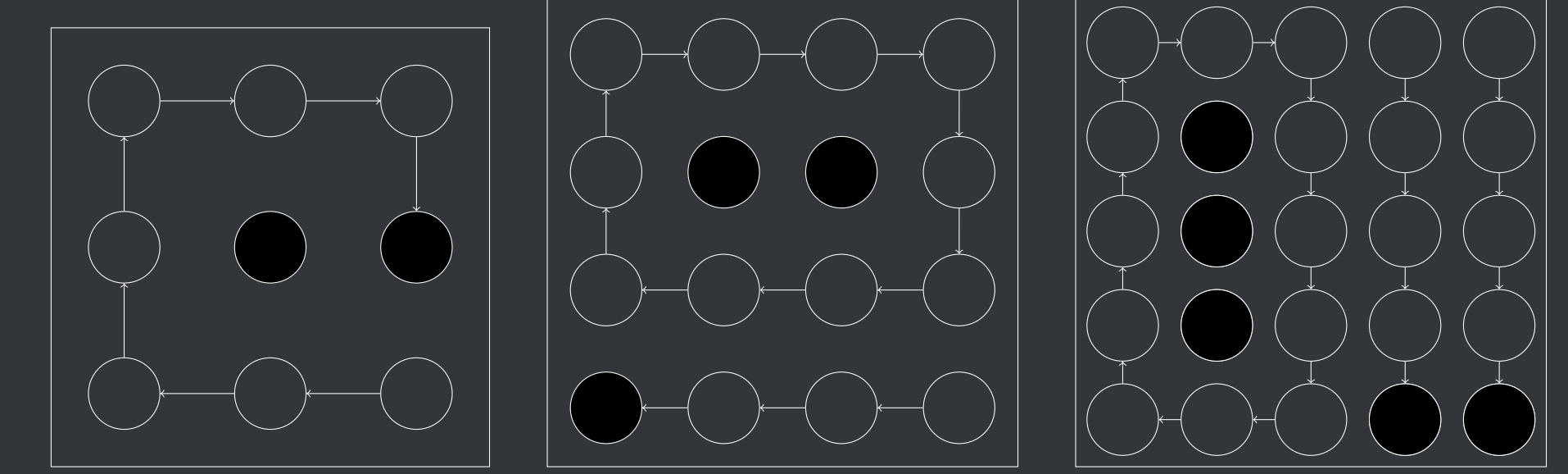
is a graphical translation.

More generally, a graph may admit several graphical translations  $f$  with various sizes of  $\mathcal{V}_{0,f}$ . When applicable, we refer to them as  $c$ -graphical translations where  $c = |\mathcal{V}_{0,f}|$ .

**Proposition :** Graphical translations on such noncyclic grid graphs are geometrical translations and  $f_\omega$

## 6. Counter Examples

On small graphs, one can find applications that verify our definitions without being geometrical translations



## Demonstrations

### Lemma (*Contamination lemma*)

Let  $f$  be a perfect graphical translation on a cyclic grid graph  $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$  with parameters  $\mathbf{d}$  and  $(\ell_i)_{1 \leq i \leq d}$ . Let  $\mathbf{v}$  be in  $\mathcal{V}$  and consider  $\delta = f(\mathbf{v}) - \mathbf{v}$ , then  $\forall \mathbf{w} \in \mathcal{V}$  neighbor of  $\mathbf{v}$ ,  $f(\mathbf{w}) = \mathbf{w} + \delta$ .

### Proof for noncyclic grid grid graphs

- Let  $f$  be a candidate graphical translation of any graph  $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$ . For any vertex  $\mathbf{v} \in \mathcal{V}$ , the sequence  $(f^n(\mathbf{v}))_n$  is either periodic or finite (in which case the last element is  $\omega$ ). We call grid graph slice  $(a, i)^\perp$  a set of vertices that share one coordinate of value  $a$  at dimension  $i$ .
- Let us consider a noncyclic grid graph  $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle$  with parameters  $\mathbf{d}$  and  $(\ell_i)_{1 \leq i \leq d}$ , and  $f$  a  $c$ -graphical translation with the largest  $c$ . Then we have  $|\mathcal{V}_{0,f}| \geq (\ell_1 - 1) \prod_{k=2}^d \ell_k$ .
- Consider a union of two adjacent slices  $\mathcal{S}$  of a noncyclic grid graph  $\mathcal{G}$ . Consider  $f$  to be a  $c$ -graphical translation with the largest  $c$ . If there exists some vertex  $\mathbf{v} \in \mathcal{V}$  such that  $(f^n(\mathbf{v}))_n$  is periodic, then  $\mathcal{S} \not\subset \mathcal{V}_{0,f}$ .
- Let us consider  $f$  to be a  $c$ -graphical translation with the largest  $c$ . Then there exists  $m$  such that  $(m, 1)^\perp \subset \mathcal{V}_{0,f}$  and  $(m+1, 1)^\perp \subset \mathcal{V}_{0,f}$ . Moreover,  $f((m, 1)^\perp \cup (m+1, 1)^\perp) \not\subset (m, 1)^\perp \cup (m+1, 1)^\perp$ .
- Let us consider  $f$  to be a  $c$ -graphical translation with the largest  $c$ . Then  $f$  is the geometrical translation by  $\mathbf{e}_1$  or  $-\mathbf{e}_1$ .

## Conclusions

- Success for cyclic grid graphs and subfamilies of noncyclic grid graphs.
- Extensions for all graphs could be found by adapting certain properties.
- Combinatorial issues to effectively find translations on a given graph.

## References

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